

Project: Automated Backup and Recovery System

Services: EC2, EBS, Lambda, CloudWatch Events

Description: Automate EBS snapshot creation and deletion using Lambda triggered by CloudWatch Events. Implement snapshot retention policies.

Skills: Automation, backup strategies, Lambda scripting.

Definition: The Automated Backup and Recovery System is designed to provide reliable and fully automated backups of AWS EC2 instance volumes (EBS). It uses AWS Lambda to create snapshots of volumes selected by specific tags and automatically deletes older snapshots based on a defined retention policy. CloudWatch Events schedules the Lambda function at regular intervals, enabling seamless, automatic backup processes.

Scope of the Project: The scope of this project is to design and implement a fully automated backup and recovery system for AWS EC2 instance volumes (EBS). The system leverages AWS Lambda and CloudWatch Events to automate the creation of snapshots and deletion of older snapshots based on a defined retention policy.

This project covers:

- Automated EBS Snapshot Creation: Automatically creates backups of all EBS volumes in the account.
- Snapshot Retention Management: Deletes older snapshots according to the retention policy to optimize storage and control costs.
- Scheduled Automation: Backups are triggered at regular intervals using CloudWatch Events, eliminating the need for manual intervention.

Methodology:

1.Lambda + CloudWatch Events - Best Method

- Uses all required AWS services: EC2, EBS, Lambda, and CloudWatch Events.
- Provides full control over backup workflows, including custom tagging and retention policies.
- Fully automated: snapshots are created and deleted without manual intervention.
- Production-ready and suitable for organizational deployment.
- Cost-effective, Free-tier friendly for small workloads
- Requires Lambda scripting and CloudWatch Event scheduling, but provides maximum flexibility and control.

2.EC2 + CRON + AWS CLI – Alternative Method

- Uses EC2 instances, EBS, CRON scheduler, and AWS CLI for backup automation.
- Enables full customization of backup logic, snapshot tagging, naming and retention policies.
- Automates backups using cron jobs and shell or Python scripts.
- Requires a continuously running EC2 instance, along with OS patching, monitoring, and security management.
- Involves higher operational effort and cost compared to serverless solutions.
- Best suited for legacy systems or use cases requiring persistent compute and deep customization.

AWS Services used in the Project:

1.Amazon EC2 (Elastic Compute Cloud)

- **Definition:** Amazon EC2 is a web service that provides resizable virtual servers in the cloud, allowing users to run applications and workloads on demand.

- **Purpose:** Provides virtual servers (instances) where applications run and EBS volumes are attached.
- **Role:** The root and attached EBS volumes of EC2 instances are the primary targets for snapshot creation.

2.Amazon EBS (Elastic Block Store)

- **Definition:** Amazon EBS is a managed block storage service designed to be used with EC2 instances, offering persistent and reliable storage.
- **Purpose:** Provides persistent block storage for EC2 instances.
- **Role:** Volumes are backed up by taking snapshots.

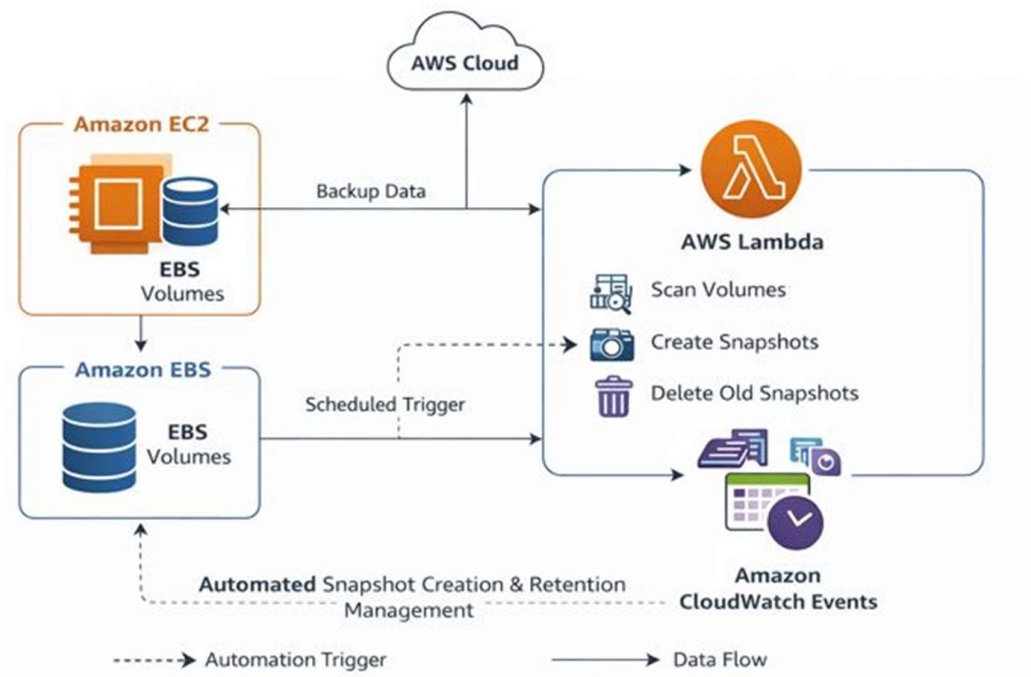
3.AWS Lambda

- **Definition:** AWS Lambda is a serverless compute service that allows users to run code without provisioning or managing servers.
- **Purpose:** Runs serverless code to automate backup operations.
- **Role:** Scans EC2 volumes, Creates snapshots of EBS volumes., Deletes old snapshots based on retention policy.

4.Amazon CloudWatch Events

- **Definition:** Amazon CloudWatch Events (now part of Amazon Event Bridge) is a service that delivers a near real-time stream of system events and enables automated actions based on schedules or events.
- **Purpose:** Schedules and triggers automated tasks.
- **Role:** Triggers the Lambda function at regular intervals to ensure snapshots are created and old snapshots deleted without manual intervention.

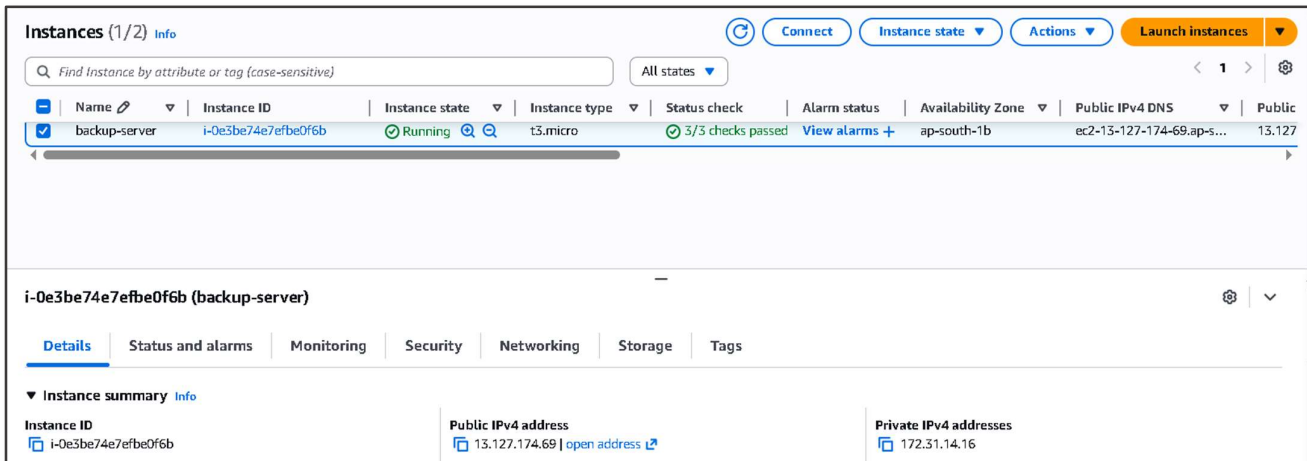
Architecture Diagram:



Implementation Steps to Do the Project:

Step 1: Launch EC2 Instance

- Go to AWS Console → EC2 → Launch Instances.
- Choose Amazon Linux.
- Configure: Default VPC, Default Subnet, Auto-assign Public IP.
- Add Root Volume: 8 GB, gp2.
- Configure Security Group
- Review & Launch.



Instances (1/2) Info

Find instance by attribute or tag (case-sensitive) All states < 1 > ⚙

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
backup-server	i-0e3be74e7efbe0f6b	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-13-127-174-69.ap-s...	13.127

i-0e3be74e7efbe0f6b (backup-server)

Details Status and alarms Monitoring Security Networking Storage Tags

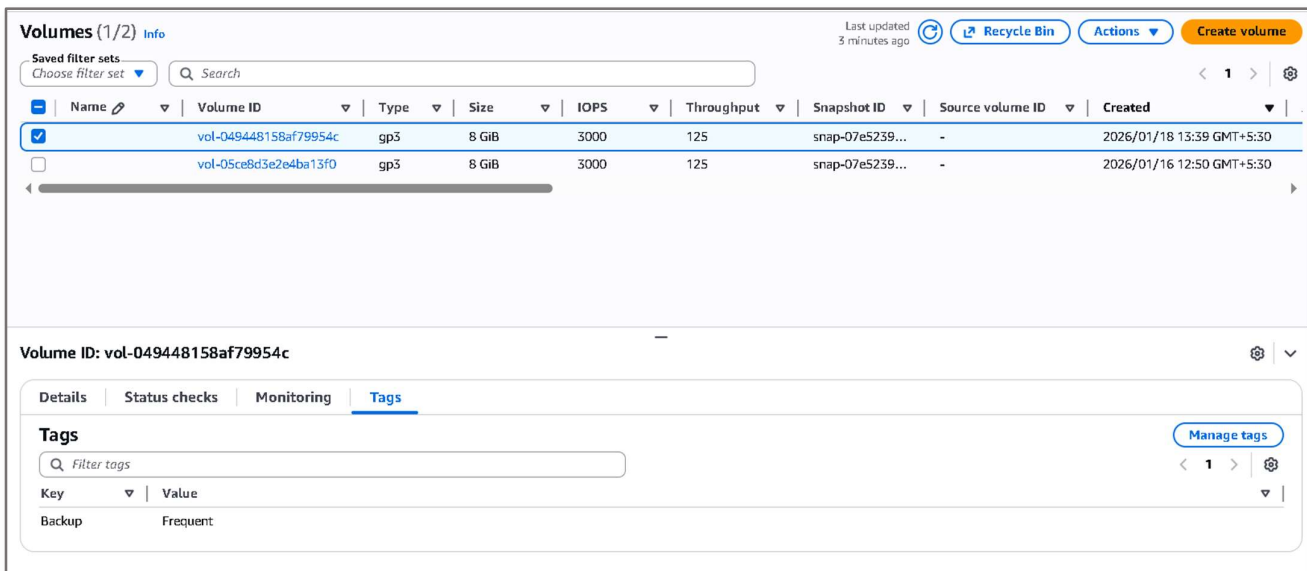
▼ Instance summary Info

Instance ID: i-0e3be74e7efbe0f6b

Public IPv4 address: 13.127.174.69 | [open address](#)

Private IPv4 addresses: 172.31.14.16

- Add Tag to volume: Key=Backup, Value=Frequent.



Volumes (1/2) Info

Save filter sets Choose filter set Search < 1 > ⚙

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created
☒	vol-049448158af79954c	gp3	8 GiB	3000	125	snap-07e5239...	-	2026/01/18 13:39 GMT+5:30
☐	vol-05ce8d3e2e4ba13f0	gp3	8 GiB	3000	125	snap-07e5239...	-	2026/01/16 12:50 GMT+5:30

Volume ID: vol-049448158af79954c

Details Status checks Monitoring Tags

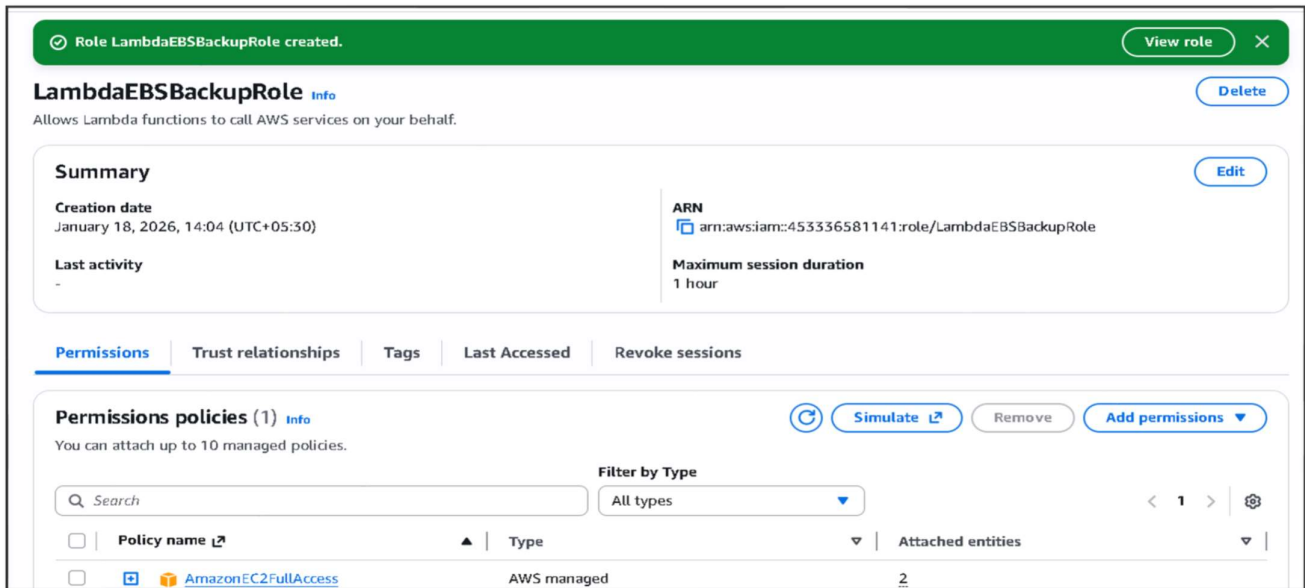
Tags

Filter tags < 1 > ⚙

Key	Value
Backup	Frequent

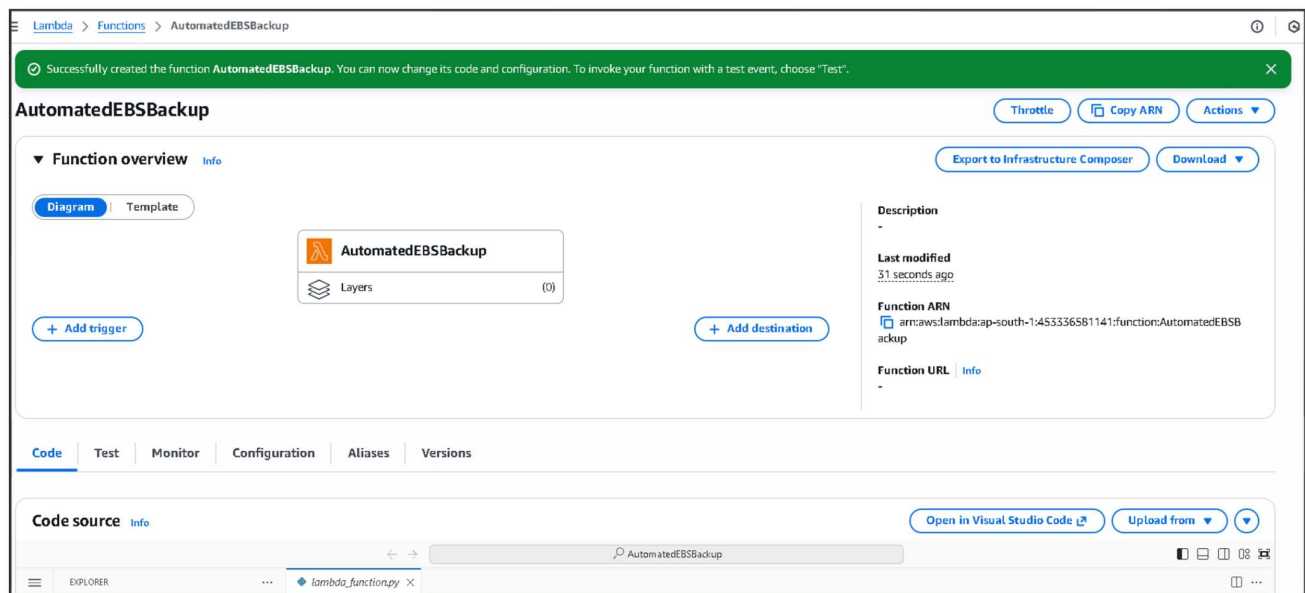
Step 2: Create IAM Role for Lambda

- Go to IAM → Roles → Create Role → Lambda.
- Attach policy: AmazonEC2FullAccess
- Name the role: LambdaEBSBackupRole



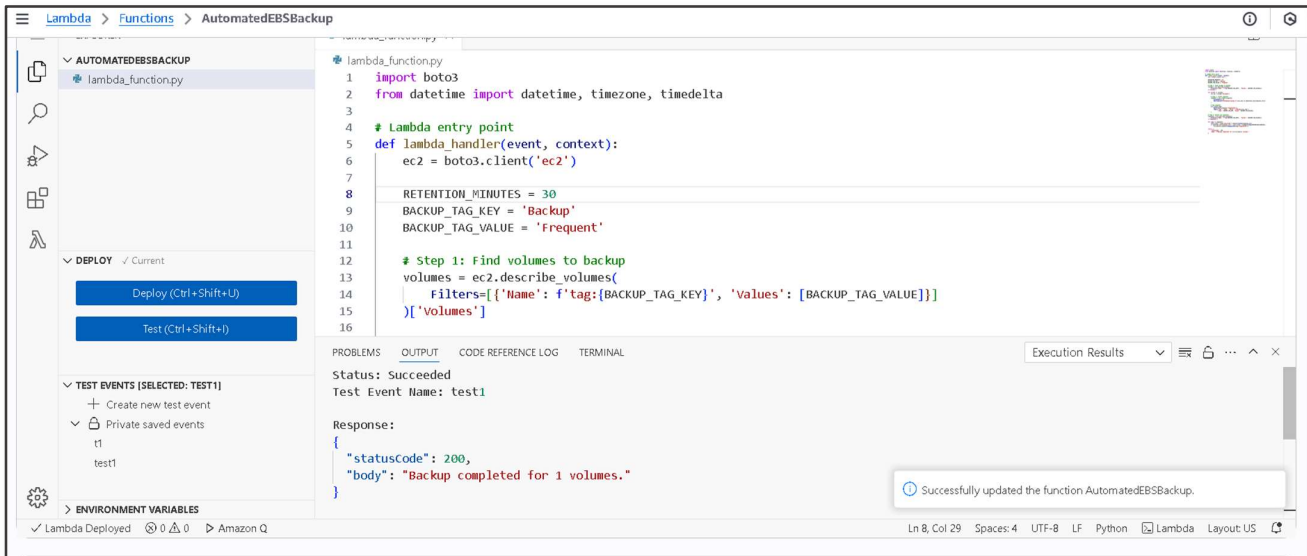
Step 3: Create Lambda Function

- Go to Lambda → Create Function → Author from Scratch.
- Function Name: AutomatedEBSBackup.
- Runtime: Python 3.10.
- Add Existing Role: LambdaEBSBackupRole.

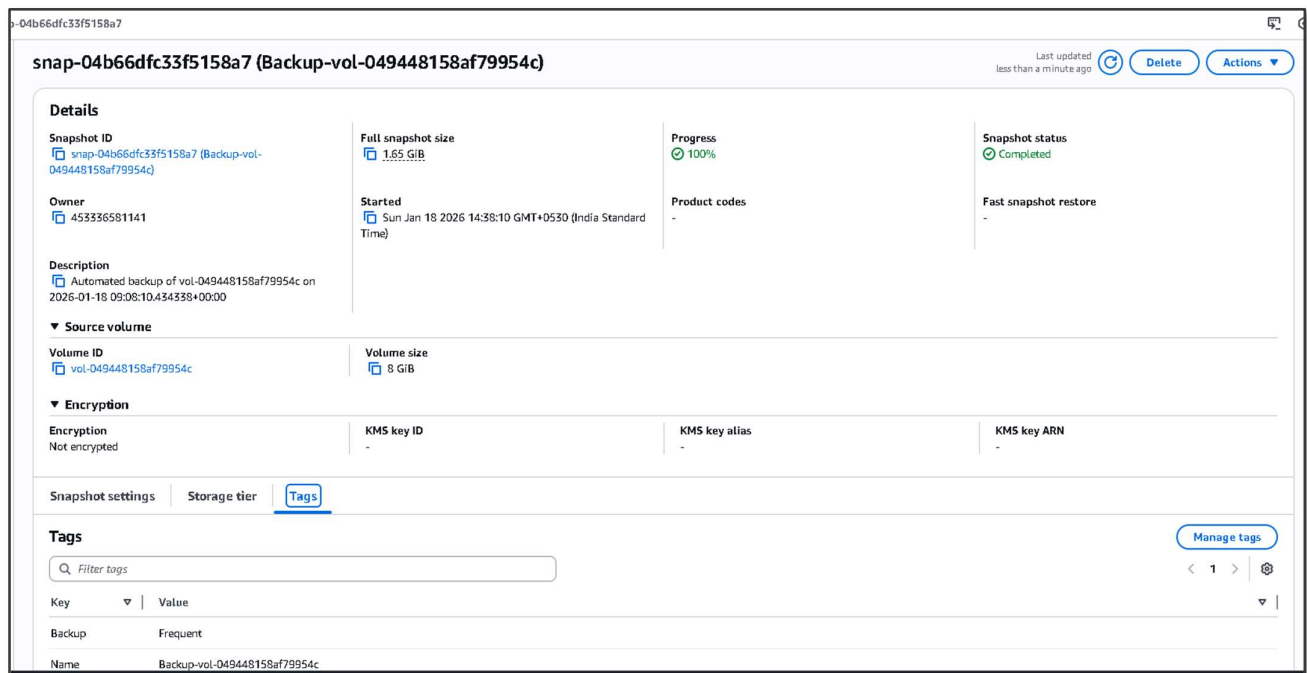


Step 4: Lambda Function code and Test

- Add the backup script
- In Lambda console, click Test → Create Test Event.
- Enter Event Name as test1.
- Click Test to run the Lambda function.



- Go to EC2 → Snapshots to confirm snapshot creation.
- Verify tags and description on the snapshot.



Step 5: Schedule Lambda Using CloudWatch Events

- Go to Event Bridge (CloudWatch Events)
- Event Source: Select Schedule → Recurring Schedule
- Schedule Pattern: Choose Cron expression
- Enter: cron (0/10 * * ? *)
- This triggers the Lambda every 10 minutes.
- Target: Add Lambda Function → AutomatedEBSBackup
- Enable Schedule
- Create Schedule

les > AutomatedEBSBackupSchedule

✓ Your schedule AutomatedEBSBackupSchedule is being created. ✕

AutomatedEBSBackupSchedule

[Disable](#) [Edit](#) [Delete](#)

Schedule detail

Schedule name
AutomatedEBSBackupSchedule

Description
Triggers Lambda function for automated EBS Backups

Schedule group name
default

Status
✓ Enabled

Schedule ARN
arn:aws:scheduler:ap-south-1:453336581141:schedule/default/AutomatedEBSBackupSchedule

Action after completion
NONE

Schedule start time
-

Schedule end time
-

Execution time zone
Asia/Kolkata

Flexible time window
-

Created date
Jan 18, 2026, 15:37:50 (UTC+05:30)

Last modified date
Jan 18, 2026, 15:37:50 (UTC+05:30)

[Schedule](#) [Target](#) [Retry policy](#) [Dead-letter queue](#) [Encryption](#)

Schedule

Cron expression [Info](#)

0/10 * * * ? *

Step 6: Verify Snapshots & Retention

- Go to EC2 → Snapshots
- Automatically snapshot of volume for every 10 minutes.

Snapshots (4) [Info](#)

Owned by me

Last updated 2 minutes ago [Recycle Bin](#) [Actions](#) [Create snapshot](#)

☐	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage t...	Snapshot status	Started	Progress
☐	Backup-vol-04...	snap-0d675415df56821be	1.65 GiB	8 GiB	Automated backup of vol-0...	Standard	Completed	2026/01/18 15:49 GMT+5:30	100%
☐	Backup-vol-04...	snap-0d7f4bc9941397022	1.65 GiB	8 GiB	Automated backup of vol-0...	Standard	Completed	2026/01/18 15:40 GMT+5:30	100%
☐	Backup-vol-04...	snap-04b66dcfc33f5158a7	1.65 GiB	8 GiB	Automated backup of vol-0...	Standard	Completed	2026/01/18 14:38 GMT+5:30	100%
☐	Backup-vol-04...	snap-054bcff9022f0d01	1.65 GiB	8 GiB	Automated backup of vol-0...	Standard	Completed	2026/01/18 15:50 GMT+5:30	100%

- Snapshots older than retention period are deleted automatically.

Snapshots (1) [Info](#)

Owned by me

Last updated less than a minute ago [Recycle Bin](#) [Actions](#) [Create snapshot](#)

☐	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage t...	Snapshot status	Started	Progress
☐	Backup-vol-04...	snap-01a07e7f73cd60c7	1.65 GiB	8 GiB	Automated backup of vol-0...	Standard	Completed	2026/01/18 17:00 GMT+5:30	100%

Troubleshooting:

Issue 1: Lambda Test Failed Due to Timeout.

The default Lambda timeout is 3 seconds, which is too short for the function to scan volumes, create snapshots, and delete old snapshots.

Solution:

Increase the Lambda function timeout.

- Go to Lambda → Your Function → Configuration.
- Click Edit.
- Set Timeout to 5 minutes.
- Save the changes.

Issue 2: Lambda Handler Not Found

Lambda could not find the entry-point function because it was not defined.

Every Lambda function requires a handler function to start execution.

Solution:

Defined a Lambda handler function in code.

Conclusion:

The Automated Backup and Recovery System provide a reliable, automated solution for protecting EC2 data through scheduled EBS snapshots. Using Lambda and CloudWatch Events, backups are created and old snapshots are automatically deleted according to the retention policy, reducing manual effort and storage costs.

The Lambda-based backup solution is scalable, flexible, and also can be applied across multiple instances or regions. It leverages AWS services efficiently, is suitable for organizational deployment, and can be implemented within the free-tier limits for small workloads.